EASY · ADVANCED MATH · ANSWER KEY

Advanced Math

05

10 answers · Rationale-rich · Teach from doc

ANSWER KEY · SAT QUESTION BANK

2026-05-29

Q1 A

A. 0, -6

Choice A is correct. The y-intercept of a graph in the xy-plane is the point x, y on the graph where $x = 0$. For the graph shown, at $x = 0$, the corresponding value of y is -6. Therefore, the y-intercept of the graph shown is $0, -6$.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q2 A

A. $W = Pt$

Choice A is correct. Multiplying both sides of the equation by t yields $P \cdot t = (\frac{W}{t}) \cdot t$, or $Pt = W$, which expresses W in terms of P and t. This is equivalent to W = Pt.

Choices B, C, and D are incorrect. Each of the expressions given in these answer choices gives W in terms of P and t but doesn't maintain the given relationship between W, P, and t. These expressions may result from performing different operations with t on each side of the equation. In choice B, W has been multiplied by t, and P has been divided by t. In choice C, W has been multiplied by t, and the quotient of P divided by t has been reciprocated. In choice D, W has been multiplied by t, and P has been added to t. However, in order to maintain the relationship between the variables in the given equation, the same operation must be performed with t on each side of the equation.

Q3 D

D. $3x^2 + 7x - 8$

Choice D is correct. The given expression is equivalent to $2x^2 + x + -9 + x^2 + 6x + 1$, which can be rewritten as $2x^2 + x^2 + x + 6x + -9 + 1$. Adding like terms in this expression yields $3x^2 + 7x + -8$, or $3x^2 + 7x - 8$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q4 B

B. The estimated number of marine mammals in the area was $1{,}800$ when the study began.

Choice B is correct. The function P gives the estimated number of marine mammals in a certain area, where t is the number of years since a study began. Since the value of $P0$ is the value of Pt when $t = 0$, it follows that $P0 = 1{,}800$ means that the value of Pt is $1{,}800$ when $t = 0$. Since t is the number of years since the study began, it follows that $t = 0$ is 0 years since the study began, or when the study began. Therefore, the best interpretation of $P0 = 1{,}800$ in this context is the estimated number of marine mammals in the area was 1,800 when the study began.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q5

B

B. $9, 328$

Choice B is correct. The point (x, y) at which the graphs of the given equations intersect is the solution to the system of equations. Subtracting 5 from both sides of the equation $x + 5 = 14$ yields $x = 9$. Substituting 9 for x in the equation $y = 4x^2 + 4$ yields $y = 4(9)^2 + 4$, or $y = 328$. Therefore, the graphs of the equations in the given system intersect at the point $(9, 328)$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6

C

C. $2x^6 - x^4 - 6x^2$

Choice C is correct. Using the distributive property to multiply the terms in the parentheses yields $(2x^3 \cdot x^3) + (2x^3 \cdot -2x) + (3x \cdot x^3) + (3x \cdot -2x)$, which is equivalent to $2x^6 - 4x^4 + 3x^4 - 6x^2$. Combining like terms results in $2x^6 - x^4 - 6x^2$.

Choices A and D are incorrect and may result from conceptual errors when multiplying the terms in the given expression. Choice B is incorrect and may result from adding, instead of multiplying, $(2x^3 + 3x)$ and $(x^3 - 2x)$.

Q7

C

C. 20,000

Choice C is correct. It's given that x represents years after 2010. Therefore, 2010 is represented by $x = 0$. On the model shown, the point with an x-coordinate of 0 has a y-coordinate of 20,000. Thus, the model estimates that in 2010, the city had 20,000 residents.

Choice A is incorrect. This is the value of x that represents the year 2010.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is approximately the number of residents the model estimates the city had in 2014, not 2010.

Q8

B

B. 2

Choice B is correct. Subtracting x^2 from both sides of the given equation yields $x^2 - 4 = 0$. Adding 4 to both sides of the equation gives $x^2 = 4$. Taking the square root of both sides of the equation gives $x = 2$ or $x = -2$. Therefore, $x = 2$ is one solution to the original equation.

Alternative approach: Subtracting x^2 from both sides of the given equation yields $x^2 - 4 = 0$. Factoring this equation gives $x^2 - 4 = (x + 2)(x - 2) = 0$, such that $x = 2$ or $x = -2$. Therefore, $x = 2$ is one solution to the original equation.

Choices A, C, and D are incorrect and may be the result of computation errors.

Q9 D

D. $15x + 5y$

Choice D is correct. Combining like terms in the given expression yields $9x + 6x + 2y + 3y$, or $15x + 5y$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q10 .5

Accept also: 1/2

The correct answer is $\frac{1}{2}$. The value of $h2$ is the value of hx when $x = 2$. Substituting 2 for x in the given equation yields $h2 = \frac{8}{52+6}$, which is equivalent to $h2 = \frac{8}{16}$, or $h2 = \frac{1}{2}$. Therefore, the value of $h2$ is $\frac{1}{2}$. Note that 1/2 and .5 are examples of ways to enter a correct answer.

Notes

Accept equivalent forms (e.g. `0.25 ≡ 1/4`). For grid-in items, decimal and fraction answers are both valid.

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